

DIRECTIONS for questions 1 to 5: Answer these questions on the basis of the information given below.

Exactly six countries, A through F, participated in a conference and each country sent at least one professor from each of five different fields – Botany, Zoology, Sociology, Psychology and Cosmology. Further, each professor belongs to exactly one field. The following table provides partial information about the number of professors sent by each country from each field:

Field Country	Botany	Zoology	Sociology	Psychology	Cosmology
A		4	3	1	6
B			9	4	2
C	5				
D		3	5		4
E	3			2	8
F		6			1

It is also known that

- i. for any country, the number of professors that it sent from each field is distinct, and for any field, the number of professors sent from that field by each country is distinct.
- ii. the total number of Botany professors at the conference is one more than the total number of professors sent by A.
- iii. the total number of Zoology professors and the total number of Psychology professors at the conference are equal and the same as the total number of professors sent by B.
- iv. the total number of professors sent by C was 11 more than that sent by B.
- v. the total number of Psychology professors is half the total number of Cosmology professors, which, in turn, is seven more than the total number of professors sent by F.
- vi. the total number of professors sent by A is ten more than the number of Sociology professors sent by F, while the number of Sociology professors sent by E is the same as the number of Botany professors sent by A.

Q1. DIRECTIONS *for question 1:* Type in your answer in the input box provided below the question.

How many Cosmology professors did C send?

Q2. DIRECTIONS *for questions 2 and 3:* Select the correct alternative from the given choices.

Which country sent the least number of Zoology professors?

☐ a) **B**

☐ b) E

☐ c) **D**

☐ d) C

Q3. DIRECTIONS *for questions 2 and 3:* Select the correct alternative from the given choices.
Which country sent the maximum total number of professors?

☐ a) **F**

☐ b) C

☐ c) **E**

☐ d) A

Q4. DIRECTIONS *for question 4: Type in your answer in the input box provided below the question.*

How many Botany professors did A send?

Q5. DIRECTIONS *for question 5:* Select the correct alternative from the given choices.

How many Sociology professors were sent by the six countries combined?

☐ a) **22**

☐ b) 52

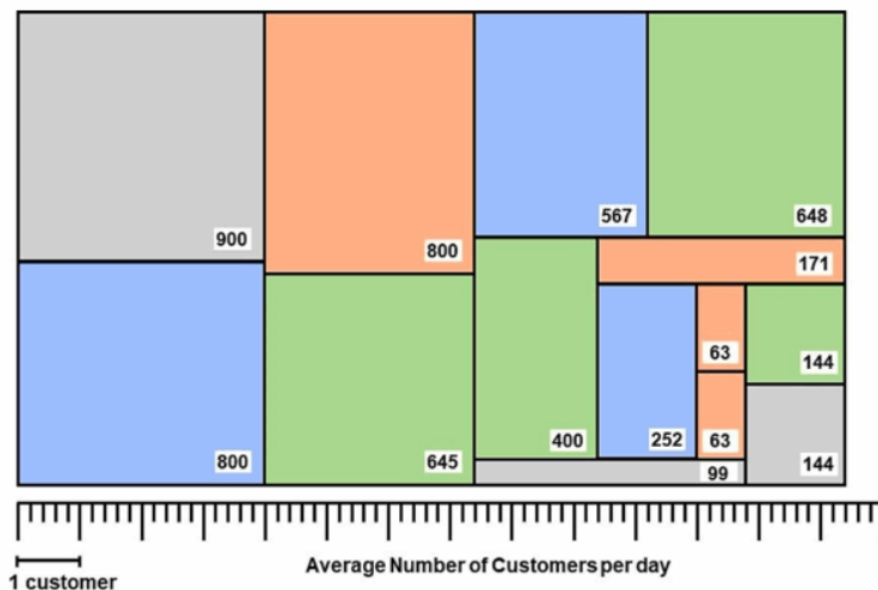
☐ c) **42**

☐ d) 32

DIRECTIONS for questions 6 to 10: Answer these questions on the basis of the information given below.

Targo, a retail chain, has stores in four different regions of the country – East, West, North and South. In the graph given below (drawn to scale), each rectangle represents each store. The area of the rectangle represents the annual revenue (in ₹ '000) of that store for the year 2022, which is mentioned in each rectangle. The width of the rectangle (along the horizontal axis) provides the average number of customers who visited that store per day during 2022 (considering all the 365 days of the year). The distance between two consecutive major gridlines along the horizontal axis represents 1 customer. The rectangles representing the stores that belong to the same region are shaded in the same colour.

Further, the revenue of any store comprises only the sales made in that store to the customers who visited that store. Also, during the year, any customer visited exactly one store and visited that store exactly once. Any store is open on all the days of the year. The average revenue per customer (ARPC) of any store is defined as the ratio of the total revenue of a store in a year to the total number of customers who visited that store in a year.



It is also known that, for 2022,

- the store with the second highest annual revenue in North had the same ARPC as the store with the least annual revenue in West.
- among all the stores across the four regions, the store with the highest ARPC was in South.

Q6. DIRECTIONS *for questions 6 and 7:* Type in your answer in the input box provided below the question.

What is the highest annual revenue (in ₹ '000) of any store in the North?

Q7. DIRECTIONS *for questions 6 and 7: Type in your answer in the input box provided below the question.*

How many stores have ARPC greater than ₹240?

Q8. DIRECTIONS *for questions 8 to 10:* Select the correct alternative from the given choices.

What is the difference between the total number of customers who visited any store in the North and that who visited any store in the East?

- ☐ a) 219
- ☐ b) 73
- ☐ c) 730
- ☒ d) 584 ✓ **Your answer is correct**

Q9. DIRECTIONS *for questions 8 to 10:* Select the correct alternative from the given choices.

What is the highest ARPC of any store in the country, approximately?

☐ a) ₹783

☐ b) ₹645

☐ c) ₹725

☐ d) ₹813

Q10. DIRECTIONS *for questions 8 to 10:* Select the correct alternative from the given choices.

For which region is the total number of customers who visited any store in that region the highest?

- ☐ a) **East**
- ☒ b) **West** ✓ **Your answer is correct**
- ☐ c) **South**
- ☐ d) **North**

DIRECTIONS *for questions 11 to 15:* Answer these questions on the basis of the information given below.

Each of four persons, Amit, Kishore, Lalit and Naveen, live in a different city among Chennai, Hyderabad, Mumbai and Bangalore. Further, each person has a different number of kids and the number of kids with any person is a distinct integer greater than 0 and less than 5.

Three persons, A, B and C, knew the cities in which each of the four persons (mentioned above) lives and the number of kids that each person has. Each of these three persons, A, B and C, made exactly three statements about the four persons mentioned above. The nine statements were made one after the other, in a certain order, such that no two consecutive statements were made by the same person. Some of the statements made by these three persons were true, while the others were false.

However, when making the statements,

- A makes any statement only when the total number of true statements made before A makes his statement is at least one and is the same as the total number of false statements that were made.
- B makes any statement only when the total number of true statements made before B makes his statement is an odd number.
- C makes any statement only when the total number of false statements that were made before C makes his statement is an even number.

The following table provides the statements made by the three persons, in the order in which they were made by that person:

Statement No.	A
1	The person with two kids lives in Hyderabad.
2	The person with three kids lives in Mumbai.
3	The person with one kid lives in Bangalore.

Statement No.	B
1	Lalit does not live in Bangalore.
2	Naveen has three kids.
3	The person who lives in Hyderabad has at least three kids.

Statement No.	C
1	Kishore has three kids.
2	Amit does not live in Chennai and he has more than one kid.
3	The person in Mumbai has a greater number of kids than the one in Chennai.

Q11. DIRECTIONS *for question 11:* Select the correct alternative from the given choices.

What BEST can be said about the city in which Kishore lives?

- ☐ a) **Chennai**
- ☐ b) **Hyderabad**
- ☐ c) **Mumbai**
- ☐ d) **Either Mumbai or Hyderabad**

Q12. DIRECTIONS *for question 12:* Type in your answer in the input box provided below the question.

How many true statements were made by the three persons combined?

Q13. DIRECTIONS *for questions 13 to 15:* Select the correct alternative from the given choices.

How many persons did not make even one true statement?

☐ a) 0

☐ b) 1

☐ c) 2

☐ d) 3

Q14. DIRECTIONS *for questions 13 to 15:* Select the correct alternative from the given choices.

How many kids does Naveen have?

☐ a) 3

☐ b) 2

☐ c) 1

☐ d) 4

Q15. DIRECTIONS *for questions 13 to 15:* Select the correct alternative from the given choices.

If the nine statements are numbered from 1 to 9, in the order in which they were made, which of the following statements are false?

I. Statement 7

II. Statement 8

III. Statement 9

- ☐ a) **Only I**
- ☐ b) **Only II**
- ☐ c) **Only I and II**
- ☐ d) **Only I and III**

DIRECTIONS *for questions 16 to 20:* Answer these questions on the basis of the information given below.

Seven lawyers – A through G – had hearings scheduled for six cases – C1 through C6 – on a particular day. For each case, one of the seven lawyers was the Prosecution Lawyer, while another was the Defence Lawyer. For each of the six cases, the judgements were passed on the same day and each case was won by one of the two lawyers of that case (i.e., either the Prosecution or the Defence). Further, the hearings of the six cases began at 10:00 am, 11:20 am, 11:50 am, 1:30 pm, 2:40 pm, and 4:20 pm, not necessarily in that order from C1 to C6. Further, for any lawyer, there was a gap of at least 120 minutes between the beginning of one hearing and the beginning of the next.

It is also known that

- i. D was a Prosecution Lawyer for exactly two cases, while F had at least two hearings scheduled after 12 noon.
- ii. A and B were the Prosecution Lawyer and Defence Lawyer, respectively, for the case scheduled at 11:20 am, but this case was neither C1 nor C5.
- iii. C, who had exactly three hearings on that day, won exactly two cases and of these two cases, he was the Prosecution Lawyer for at least one case.
- iv. E was the Defence Lawyer for only one case, which is C2, and this was the only case won by D.
- v. the cases whose hearings began at 10:00 am, 11:20 am and 4:20 pm were won by the Defence Lawyers of those cases, while the cases C2 and C4 were won by the Prosecution Lawyers.
- vi. G was one of the lawyers for C3, whose hearing was scheduled at 11:50, while E won at least one case.

Q16. DIRECTIONS *for questions 16 to 20:* Select the correct alternative from the given choices.

How many of the seven lawyers did not lose any case on that day?

☐ a) 4

☐ b) 3

☐ c) 1

☐ d) 5

Q17. DIRECTIONS *for questions 16 to 20:* Select the correct alternative from the given choices.

For which of the following cases was C the Defence Lawyer?

- ☐ a) **Either C3 or C5**
- ☐ b) **C4**
- ☐ c) **Either C1 or C4**
- ☐ d) **Either C1 or C5**

Q18. DIRECTIONS *for questions 16 to 20:* Select the correct alternative from the given choices.

Against which of the following lawyers did D lose a case?

- ☐ a) **C**
- ☐ b) **G**
- ☐ c) **Both C and G**
- ☐ d) **F**

Q19. DIRECTIONS *for questions 16 to 20:* Select the correct alternative from the given choices.

For which of the following cases was E the Prosecution Lawyer?

- ☐ a) **C1**
- ☐ b) **C3**
- ☐ c) **Either C3 or C5**
- ☐ d) **C5**

Q20. DIRECTIONS *for questions 16 to 20:* Select the correct alternative from the given choices.

Which lawyer won the case C4?

☐ a) **G**

☐ b) **D**

☐ c) **C**

☐ d) **F**